HVAC and Chiller Plant Energy Efficiency Audit across Academic Blocks

Document ID: DOC_ENG_019 | **Type:** AUDIT_REPORT | **Publication:** 2024-07-22 | **Institution:** VESIT (VESIT Green Campus, Chembur)

HVAC AND CHILLER PLANT ENERGY EFFICIENCY AUDIT ACROSS ACADEMIC BLOCKS

Document Reference: VESIT/AUD/HVAC-ENERGY/2024/01

Conducting Authority: Thermal Energy Systems Research Lab in collaboration with External BEE-Accredited Energy Auditors.

Scope: Academic Block A (BLD_ACA), Academic Block B (BLD_ACB), and Computer Laboratories (BLD_LAB).

Location: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. EXECUTIVE CONTEXT AND COOLING ENERGY INVENTORY

Space air conditioning and ventilation represent the single largest operational electricity load at Vivekananda Education Society's Institute of Technology (VESIT Green Campus), accounting for 38.4% (907,160 kWh) of the total annual campus electricity consumption (2,362,400 kWh) in 2024. Operating in Mumbai's hot and humid tropical coastal climate, space cooling is critical for maintaining thermal comfort and equipment reliability across 32,500 m2 of institutional floor space.

This engineering audit evaluated cooling equipment across the primary academic blocks:

- Academic Block A (BLD_ACA, 7,500 m2): 74 split and multi-split AC units (cooling capacity: 135 TR).

- Academic Block B (BLD_ACB, 6,800 m2): 62 split AC units and ductable units (cooling capacity: 118 TR).

- Computer Laboratories (BLD_LAB, 5,200 m2): Variable refrigerant flow (VRF) systems and precision cooling units serving 16 computer labs and two server rooms (total capacity: 160 TR).

2. AUDIT FINDINGS AND OPERATIONAL INEFFICIENCIES

Detailed data logging across 240 operating hours revealed major efficiency bottlenecks:

- Sub-Optimal Operating Temperature Setpoints: Over 62% of faculty cabins and computer labs maintained thermostat setpoints between 19.0°C and 20.0°C, significantly below the Bureau of Energy Efficiency (BEE) recommended default comfort setpoint of 24.0°C. Controlled thermal testing confirmed that every 1.0°C increase in thermostat setpoint yields a measured 6.0% reduction in compressor electricity consumption.

- Condenser Coil Fouling and Particulate Accumulation: Ambient particulate matter in Chembur caused severe dust fouling on outdoor condenser coils of 14 split AC units in BLD_ACA, causing an 18.2% degradation in the equipment's Coefficient of Performance (COP) and forcing compressors to run continuously.

- Refrigerant GWP Profiles: 45 legacy split AC units continue to operate using HCFC-22 (R-22) refrigerant, which possesses a high Global Warming Potential (GWP of 1,810) and ozone-depleting properties.

3. MANDATED ACTION PLAN AND CONTROL MEASURES

The audit mandates four immediate engineering and policy interventions:

3.1 Mandatory Thermostat Lock at 24.0°C:

Effective August 1, 2024, all air conditioning thermostats across administrative offices, lecture halls, and faculty rooms were locked at 24.0°C via tamper-proof physical covers and digital central VRF controllers. Raising average operating setpoints from 20.0°C to 24.0°C yields an immediate 24.0% reduction in cooling energy draw.

3.2 Rigorous Preventative Maintenance Schedule:

Mandate bi-monthly chemical cleaning and pressure-washing of outdoor condenser coils using harvested rainwater from INIT_RWH_SYSTEM, restoring original manufacturer COP ratings.

3.3 Phase-Out of R-22 Refrigerants:

Establish a rolling procurement schedule to retire aging R-22 air conditioners by 2026, transitioning to high-efficiency inverter units utilizing low-GWP refrigerants: R-32 (GWP 675) for split units and R-290 propane (GWP 3) for specialized pilot lab applications.

4. ENERGY CONSERVATION AND CARBON ABATEMENT IMPACT

Full execution of the HVAC action plan will achieve:

- Annual Electricity Savings: 142,000 kWh/year across BLD_ACA, BLD_ACB, and BLD_LAB.

- Carbon Emissions Avoided: Applying Central Electricity Authority factor EF_ELEC_GRID (0.82 kg CO2e/kWh), the intervention eliminates 116,440 kg CO2e annually.

- Utility Bill Savings: INR 15,62,000 annually against an implementation cost of INR 3,80,000, achieving a payback period of only 2.9 months.

These results advance institutional targets TRG_NET_ZERO and TRG_RENEWABLE while supporting United Nations SDG 7, SDG 11, and SDG 13.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_ENG_019**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*